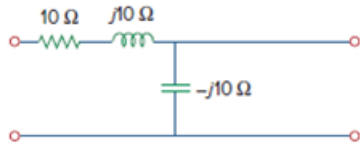


Problem

Find the z parameters of the circuit in Fig. 19.67.

Figure 19.67

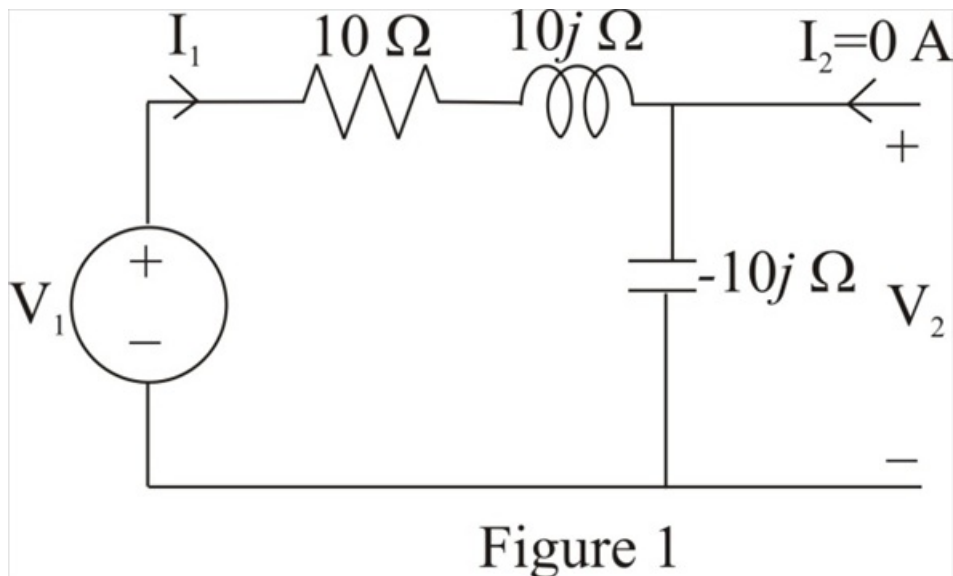


Step-by-step solution

Step 1 of 8

Refer to Figure 19.67 in the text book.

Connect the voltage source $\mathbf{V}_1$ as shown in Figure 1. Identify the currents and voltages in the circuit.



Step 2 of 8

Consider the expression for the voltage $\mathbf{V}_1$.

$$\begin{aligned}\mathbf{V}_1 &= \mathbf{I}_1(10 + j10 - j10) \\ &= \mathbf{I}_1(10)\end{aligned}$$

Therefore, the expression for the voltage $\mathbf{V}_1$ is $\mathbf{I}_1(10)$.

Consider the expression for the voltage $\mathbf{V}_2$.

$$\mathbf{V}_2 = \mathbf{I}_1(-j10)$$

Therefore, the expression for the voltage $\mathbf{V}_2$ is $\mathbf{I}_1(-j10)$.

Step 3 of 8

Determine the expression for the open circuit input impedance z_{11} .

$$z_{11} = \frac{V_1}{I_1}$$

Substitute $I_1(10)$ for V_1 .

$$\begin{aligned} z_{11} &= \frac{I_1(10)}{I_1} \\ &= 10 \Omega \end{aligned}$$

Therefore, the expression for the open circuit input impedance z_{11} is 10Ω .

Step 4 of 8

Determine the value of the open circuit transfer impedance z_{21} from port-2 to port-1.

$$z_{21} = \frac{V_2}{I_1}$$

Substitute $I_1(-j10)$ for V_2 .

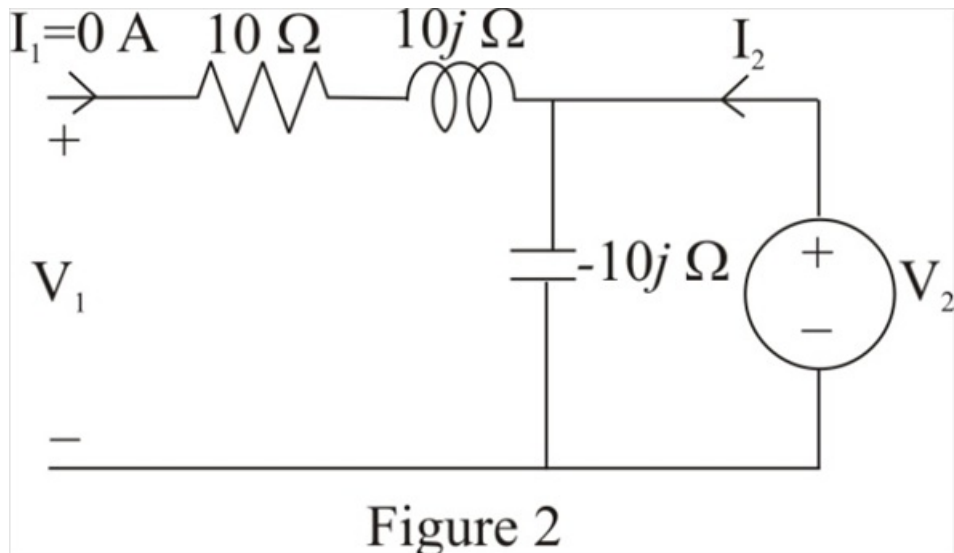
$$\begin{aligned} z_{21} &= \frac{I_1(-j10)}{I_1} \\ &= -j10 \Omega \end{aligned}$$

Therefore, the value of the open circuit transfer impedance z_{21} from port-2 to port-1 is,

$$-j10 \Omega.$$

Step 5 of 8

Connect the voltage source V_2 as shown in Figure 2. Identify the currents and voltages in the circuit.



Step 6 of 8

Consider the expression for the voltage $\mathbf{V}_2$.

$$\mathbf{V}_2 = \mathbf{I}_2(-j10)$$

Therefore, the expression for the voltage $\mathbf{V}_2$ is $\mathbf{I}_2(-j10)$.

Consider the expression for the voltage $\mathbf{V}_1$.

$$\mathbf{V}_1 = \mathbf{I}_2(-j10)$$

Therefore, the expression for the voltage $\mathbf{V}_1$ is $\mathbf{I}_2(-j10)$.

Step 7 of 8

Determine the value of the open circuit transfer impedance $\mathbf{z}_{12}$ from port-1 to port-2.

$$\mathbf{z}_{12} = \frac{\mathbf{V}_1}{\mathbf{I}_2}$$

Substitute $\mathbf{I}_2(-j10)$ for $\mathbf{V}_1$.

$$\begin{aligned}\mathbf{z}_{12} &= \frac{\mathbf{I}_2(-j10)}{\mathbf{I}_2} \\ &= -j10 \, \Omega\end{aligned}$$

Therefore, the value of the open circuit transfer impedance $\mathbf{z}_{12}$ from port-1 to port-2 is

$$\boxed{-j10 \, \Omega}.$$

Step 8 of 8

Determine the expression for the open circuit output impedance $\mathbf{z}_{22}$.

$$\mathbf{z}_{22} = \frac{\mathbf{V}_2}{\mathbf{I}_2}$$

Substitute $\mathbf{I}_2(-j10)$ for $\mathbf{V}_2$.

$$\begin{aligned}\mathbf{z}_{22} &= \frac{\mathbf{I}_2(-j10)}{\mathbf{I}_2} \\ &= -j10 \, \Omega\end{aligned}$$

Therefore, the expression for the open circuit output impedance $\mathbf{z}_{22}$ is $\boxed{-j10 \, \Omega}$.